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Trends in Cognitive Sciences

Review

The interoceptive origin of reinforcement learning

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Rewards play a crucial role in sculpting all motivated behavior. Traditionally, research on reinforcement learning has centered on how rewards guide learning and decision-making. Here, we examine the origins of rewards themselves. Specifically, we discuss that the critical signal sustaining reinforcement for food is generated internally and subliminally during the process of digestion. As such, a shift in our understanding of primary rewards as an immediate sensory gratification to a state-dependent evaluation of an action's impact on vital physiological processes is called for. We integrate this perspective into a revised reinforcement learning framework that recognizes the subliminal nature of biological rewards and their dependency on internal states and goals.

Highlights

The reinforcing effects of food and fluids are driven by post-oral interoceptive primary reward signals, which reflect key physiological resources essential for sustaining life, such as energy, nutrients, and hydration.

Primary rewards are accompanied by a cascade of earlier signals – secondary and proxy rewards – that facilitate learning and prospective control rather than sustaining reinforcement.

Primary reward signals are dependent on internal states and goals.

The traditional reinforcement learning framework needs to be extended to address the generation of state-dependent reward signals and their interaction with reinforcement learning mechanisms within biological brains.

What are the origins of reward?

The concept of reward is a cornerstone across behavioral science, cognitive neuroscience, and artificial intelligence. Rewards serve as powerful incentives and essential teaching signals that guide learning in biological and artificial systems: from the tasty morsel that motivates a mouse to the monetary incentives that reinforce human behavior and the points that train algorithms to master chess. Much research, particularly in reinforcement learning (RL) [1–6], has focused on how we learn from rewards and the role of dopamine as a 'common currency' for processing rewards to reinforce a behavior [2,5,7–10] (Box 1). However, a critical question remains unanswered: where does the reward signal originate?

Traditionally, rewards have been viewed as immediate outcomes of actions, such as the amount of juice given to an animal or the points awarded in an experiment. In this view, rewards are static, quantifiable external entities. This is at odds with what we know about biological agents: first, it neglects the complexity and dynamic nature of biological systems, where no singular sensory pathway is dedicated to reward detection [11,12]. Instead, reward must be inferred on the basis of noisy sensory signals, internal state, and context [12,13]. Second, the degree to which an outcome is rewarding depends on an agent's goals. Although designing specific reward functions for different goals in artificial agents is a nontrivial problem [14–17], biological agents routinely set their own goals. Therefore, reward cannot be inherent to the stimulus but must be computed within the organism [18–23]. Notably, RL has been largely agnostic about the origin of reward signals and instead focused on how such signals shape behavior.

Recent findings from studies of ingestive behavior clearly demonstrate that the critical reinforcing signals from food and water are not linked directly to consumption [18]. Instead, they originate from delayed post-oral **primary reward** (see Glossary) signals generated during digestion, which are necessary and sufficient for reinforcing behavior. In other words, the subjective value of food or water lies in an organism's ability to transform them into vital resources – nutrients, energy, hydration, or reproductive support – that are sensed and projected to the dopamine system. The discovery of these interoceptive primary rewards allows researchers, for the first

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from what I understand, they argue that reward is not something that emerges simply after an action is taken, it has to be inferred from noisy sensory signals and the reward is inherent depending on the goals of the agent.

the argument of this review is that, we need to stop thinking of rewards as something external and static but they are internal and the intrinsic nature of rewards needs to be incorporated in RL.

and they take food consumption as an example here.

Box 1. Reinforcement learning and the dopamine system

RL models have been remarkably successful at describing how organisms learn from rewards [6]. At the heart of this approach is an agent that interacts with its environment via actions (a_t). Those actions result in changes in the environment (changes in state s_t) and in outcomes that affect the agent (rewards r_t) (see Figure 3C in the main text). RL aims to solve the problem of learning an action policy – picking an action given states and previous outcomes – that maximizes the cumulative long-term expected reward for the agent. Because agents do not necessarily know which actions lead to the best long-term outcomes, they need to learn the optimal policy at any given moment in time t. One way to find the optimal policy is via temporal-difference learning (TD), where the agent iteratively evaluates and updates its policy on the basis of differences between the actual observed rewards r_t and the expected values V – the sum of expected future rewards– by learning from reward prediction errors (RPE) δ_t:

$$\delta_t = r_t + V_t(s_{t+1}) - V_t(s_t)$$
$$V_{t+1}(s_t) = V_t(s_t) + \alpha\delta_t,$$

where α is the amount of value update or the learning rate. One of the major scientific insights of the past decades was the discovery that phasic activity in midbrain dopamine neurons seems to reflect TD-like RPEs between the received reward and the expected time and magnitude of reward that are directly associated with the reinforcement of behavior [5,10,115–117] (see Figure 2 in the main text).

The distinction between reward and value is critical in this algorithmic context. 'Reward' refers to the valenced outcome of an action that reinforces behavior. In contrast, 'value' represents the summed expectation of future rewards, integrating past experiences and the potential long-term benefits or costs of an action. In other words, value is inherently future-oriented, subjective, and model- and experience-dependent, whereas reward serves as a 'ground-truth' correction signal and is thus linked more closely to the current time point and treated as an objective 'external' feedback.

Note that RL agents, in the strict sense, do not generate their own goals, because they lack control over their reward function: reward is an immutable objective, which is externally defined and shapes behavior through optimization rather than autonomous goal-setting. To model agents capable of goal-setting, we must consider the mechanisms by which they generate their own reward functions [112]. In biological agents, even primary rewards such as food and water depend on internal states, requiring flexible valuation. Understanding this adaptability could inform models of how higher-level goals shape reward functions in other contexts.

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the difference between rewards and values.

reward is the ground thruth, what the agent recieves when it performs the action. it is present oriented.

value is future oriented. it is the expected reward computed through calculating the past reward and the expected future rewards.

time, to derive precise empirical constraints on the questions of how biological agents generate reward signals and how these signals depend on internal states. Here, we review the literature on late reinforcing signals for sugar, fat, and water to identify these constraints and sketch how this knowledge could be incorporated into formal frameworks of RL.

Primary reward signals represent essential physiological variables

Where do reward signals for food or water come from? During digestion, the energy required to support cellular functions is released from foods. Because organisms must obtain energy to sustain life, a reinforcing mechanism linking nutrient digestion with behavior offers an adaptive advantage. Because of the significant delay between consumption and nutrient digestion, it was initially thought that reinforcement from food involves hormone release that retroactively 'stamps in' reward [24]. However, although hormones can regulate reward sensitivity [25,26] and internal states [27,28], hormones themselves are not generally reinforcing [with the exception of cholecystokinin (CCK); see below]. There is no evidence that animals will work to self-administer feeding and satiety hormones (e.g., ghrelin, insulin, leptin) in the absence of consumption. Rather, hormones primarily produce an inhibitory effect on behavior (e.g., satiety and meal termination), with the exception of ghrelin, which stimulates hunger and feeding [29,30] and can act as a negative reinforcer to condition avoidance [31]. So, if not from hormones, where does the reinforcement come from? To answer this question, let us turn to the case of sugar.

The case of sugar

When a mouse licks a spout to obtain a sugar solution, dopamine levels in the striatum rise. However, seminal work over the last decade has shown that neither licking nor dopamine release is sustained if a non-nutritive sweetener (e.g., sucralose) is used instead of sugar (e.g., sucrose or glucose) [19,23,32,33]. Moreover, even sweet taste 'blind' mice exhibit sugar-conditioned preferences and dopamine release [19,33], indicating that sweetness is neither necessary nor sufficient to sustain glucose-related dopamine release and appetitive responding.

Instead, sugar reinforcement arises from the oxidation of glucose to produce ATP, which is the fundamental energy source for cells. In other words, any glucose source is reinforcing because it supplies usable energy. Supporting this, a study in mice demonstrated that although glucose ingestion elevates extracellular dopamine in the striatum, simultaneous intravenous infusion of the glucose antimetabolite 2-deoxyglucose (2DG), which impairs ATP production, suppresses both dopamine efflux and licking behavior, implying that glucose is rewarding only when it is converted into cellular energy [23]. Similarly, human fMRI studies reveal that striatal responses to calorie predictive flavor cues reflect the metabolic impact of prior consumption [i.e., increases in plasma glucose and **dietary-induced thermogenesis (DIT)**] [34,35].

Recent evidence has pointed to the location of an energy sensor in the hepatoportal vein, a blood vessel transporting nutrients from the gastrointestinal tract to the liver [36], which generates neural signals carried via the vagus nerve to the dorsal striatum (DS) (Figure 1). Direct infusion of metabolizable, but not nonmetabolizable, sugar here increases dopamine levels in the striatum and conditions flavor preference. Additionally, a sugar-sensing pathway from the upper intestine has been identified [20,33,37], although its distinct function remains unknown.

The case of fat

Dietary fat can also lead to sustained reinforcement. It is sensed by a distinct group of cells in the upper intestine, which generate a signal that is conveyed by the vagus nerve to the DS [20,37,38] (Figure 1). Animals will work to optogenetically stimulate this gut-derived neuronal reward pathway [20]. Additionally, gut administration of CCK, which is usually contingent on nutrient sensing, also activates this pathway to cause dopamine release. Like glucose, intragastric infusion of lipids results in dopamine efflux in the DS to facilitate ingestive behavior, and this effect can be blocked with bilateral vagotomy [39]. There is evidence that the sugar and fat pathways not only arise from distinct mechanisms but also project to discrete populations of fat-sensitive or sugar-sensitive neurons in the nodose ganglion, hindbrain, and midbrain, such that there are parallel but segregated reinforcing signals to the DS [38]. Concurrent activation of both pathways potentiates reinforcement in humans [40] and in mice [38] so that, calorie for calorie, foods with both fat and carbohydrate are more reinforcing (and release more dopamine) than those with fat or carbohydrate alone. This example highlights the importance of signals generated by the organism rather than characteristics inherent to the food (e.g., calories) as the key drivers of reinforcement.

The case of water

As with sugar and fat, the primary reinforcing signal for water consumption stems from post-oral feedback [41] (Figure 1). Water-deprived animals exhibit robust, time-locked dopamine responses in the ventral tegmental area (VTA) during licking, followed by two sustained post-ingestion activations in the striatum: once water enters the gastrointestinal tract and once as it is absorbed into the body. These secondary dopamine responses are also observed when water is infused directly into the stomach, bypassing oral consumption, and are absent when water is replaced by a dehydrating hypertonic solution. Silencing VTA dopamine neurons during the post-absorptive phase, which was previously associated with activation of the DS, impairs the learned acquisition of water reinforcement, indicating a sophisticated feedback system sensitive to systemic osmolality. Thus, although sugar and fat consumption are rewarding because they supply the body with energy, water consumption is rewarding because it leads to (the detection of) systemic rehydration.

Glossary

Cue-potentiated feeding: the stimulation of food consumption by cues that have become associated with food.

Dietary-induced thermogenesis (DIT): the increase in energy expenditure above basal levels observed following nutrient consumption. It reflects the energy use associated with absorbing the nutrient, which is thought to be determined primarily by the energy content of the food.

Drive: an internal state that reflects an imbalance or need and motivates an organism to engage in behavior aimed at restoring balance. Physiological drive states are often represented in an anticipatory manner, reflecting allostatic rather than homeostatic control.

Event-driven primary reward signal: a reward signal whose occurrence is causally linked to a physiological event (e.g., conversion of sugar into energy). It can drive reinforcement in the absence of a deficit (post-oral 'appetition'), although it will be more effective in a deprived state.

Interoception: the overall process of how the nervous system (central and autonomic) senses, interprets, and integrates signals originating from within the body, providing a moment-by-moment mapping of the internal landscape of the body across conscious and nonconscious levels.

Primary reward: a reward signal capable of directly reinforcing behavior by activating the brain's dopamine system without dependence on another underlying reward signal or prior learning (unconditioned stimulus).

Proxy reward: a reward signal that offers an 'early affective draft' of the expected long-term value of an outcome. The function of proxy rewards might be to help in assigning credit between actions and their delayed outcomes. Proxy rewards have immediate reinforcing effects on behavior that do not necessarily require learning, but, similar to secondary rewards, they cannot sustain reinforcement without the presence of a primary reward signal.

Secondary reward: a learned reward signal that is reinforcing because it predicts the occurrence of a primary reward (conditioned stimulus). It cannot sustain reinforcement on its own without the presence of a primary reward.

In summary, the reinforcing effects of foods and water are driven by distinct subliminal post-oral primary reward signals. These interoceptive reward signals reflect key physiological resources essential for sustaining life (e.g., energy, nutrients, or hydration), such as the metabolism of glucose into ATP, detection of lipids in the upper gut, and systemic osmolality changes. These events are directly signaled to the brain's dopamine system, revealing a potential mechanism through which organisms optimize decision-making on the basis of the experienced nutritive value of a food or beverage.

State-driven primary reward signal: a reward signal whose occurrence is causally linked to a beneficial change in state (i.e., to maintain homeostasis). It can be elicited only if a deficit is present (e.g., water is only rewarding in a state of dehydration).

The role of secondary and proxy rewards

If interoceptive primary rewards ultimately drive food reinforcement, what role do earlier sensory cues (i.e. vision, olfaction, and taste) play in guiding learning and action selection?

Proxy rewards solve the credit assignment problem

Primary reward signals arrive with a delay: the benefits from eating or drinking unfold over minutes to hours because digestion, absorption, and rehydration are slow. Despite this, they still reinforce preceding cues and behaviors [19,21,22]. This temporal gap between action (consumption) and outcome (reward) poses a 'credit assignment' problem, where organisms must link a delayed outcome to the preceding behavior [6,42].

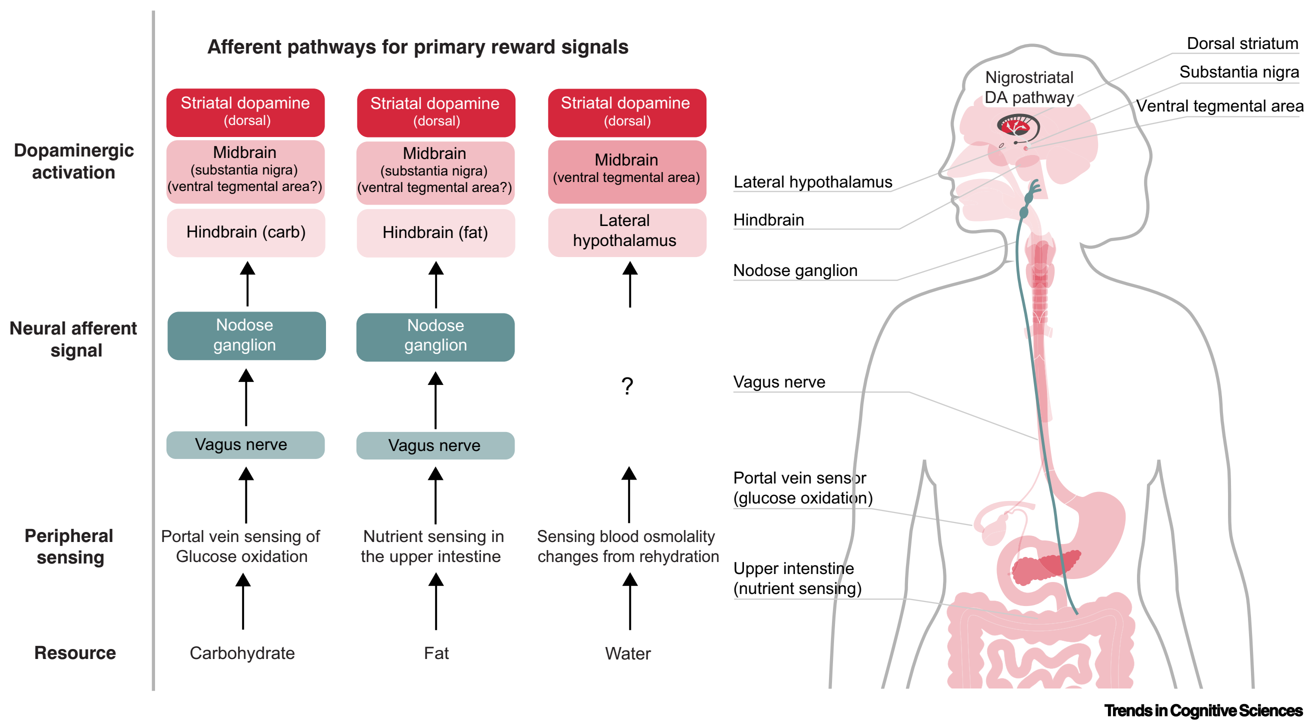

Figure 1. Distinct afferent pathways for primary reward signals through which carbohydrates, fats, and water activate the midbrain dopamine (DA) system to promote reinforcement. Carbohydrates (sugar): Although the exact afferent pathway for carbohydrates is still the subject of investigation, there is evidence that glucose oxidation after sugar consumption is sensed in the portal vein [36] and transmitted through the vagus nerve and the nodose ganglion to the midbrain dopamine system [18,54]. Fats (lipids): Fat is detected in the upper intestine, where it activates peroxisome proliferator-activated receptors (PPAR-α). This activation sends reinforcing signals via the vagus nerve to the nodose ganglion and hindbrain, which in turn increases dopamine release through the substantia nigra and the ventral tegmental area (VTA) [20,38]. Although the pathways for fat and sugar appear similar, they are parallel and activate distinct subpopulations [38]. There is also evidence that sugar is sensed in the upper intestine, leading to vagal signals [33,37]. Whether this pathway is redundant or distinct in function from the hepatoportal sensor is unknown. Water: The pathway for water is still partially unknown. However, systemic rehydration is likely detected from changes in blood osmolality and projected to dopamine neurons in the VTA via GABAergic projections from the lateral hypothalamus tracking fluid balance [41]. Although these pathways have been studied primarily in rodents, it is believed that similar mechanisms exist in humans.

Biological agents navigate this by leveraging early sensory signals that predict the expected physiological outcomes. Food and water consumption are accompanied by a cascade of early signals, including both pre-oral (e.g., vision and olfaction) and intra-oral (e.g., taste, flavor, and texture) cues, which help estimate future hydration levels and energy availability by signaling a potential nutrient or toxin presence [41,43–46].

Both pre-oral and intra-oral signals are linked to phasic dopamine activity in humans [47,48] and animals [20,33,38,39,41] and have immediate reinforcing effects on behavior. Although responses to pre-oral cues (e.g., the sight of a food) require learning (**secondary reward**), intra-oral signals – particularly taste – produce immediate and innate subjective experiences of liking/disliking (Box 2). The immediate and innate dopamine response following consumption and its effect on behavior initially led to the belief that intra-oral signals were the primary reinforcers driving conditioned responses to earlier food cues (Figure 2A). However, early reward signals alone do not sustain reinforcement or dopaminergic activity without the primary delayed reinforcement signal (Figure 2B).

This is evident in flavor-nutrient conditioning (FNC), where rodents prefer flavors linked to post-oral energy signals (e.g., glucose oxidation) over non-nutritive sweeteners with pleasant taste [21,22,49]. Disrupting the generation of these post-oral signals blocks dopamine release and FNC [23,33]. Additionally, animals that lack the ability to sense sweet taste also prefer fluids containing sugar on the basis of the presence of energy [19,23]. Thus, oral sensory signals are neither necessary nor sufficient to sustain reinforcement. In other words, they fully classify as neither primary nor secondary (learned) reward signals. We suggest reconceptualizing them as **proxy reward** signals (Box 2).

What is the role of proxy rewards? One proposal has been that innate affective responses serve to provide an 'early affective draft' of the long-term expected value, bridging the delay between

Box 2. The matter of taste

Oral signals include gustatory, retronasal, olfactory, and oral somatosensory inputs. Each is transduced by distinct receptors but integrated in the brain to produce a unitary perception of flavor [118]. Oral sensations play a privileged role in facilitating credit assignment because they pair the action (consumption) with a proxy of the expected delayed outcome, which may be subjectively experienced as liking or disliking [42,119]. It is also important to appreciate that taste and flavor perceptions play distinct roles in guiding intake.

Taste perceptions (sweet, sour, salty, bitter, and savory) serve as indicators of potential nutritional values or harmful consequences of ingested substances. However, contrary to intuition, taste plays a rather limited role in identifying foods and flavors. Rather, this depends on retronasal olfaction, which occurs when volatiles from the mouth reach the olfactory receptors in the nasal cavity and then evoke quality-specific codes in the primary olfactory cortex [120]. Consequently, we can perceive the sense of strawberry both as an aroma in the world and as a flavor in our mouths. However, we can only know how sweet that strawberry is by putting it in our mouths.

Affective responses to taste are stable: they are observed in newborns of many species [121], indicating that they are either innate or learned *in utero*. In contrast, flavor preferences are largely learned [22]. This points to an important distinction for their respective roles in reinforcement learning: flavor perception enables organisms to learn new associations between nutritional outcomes with specific foods, allowing the development of preferences or aversions based on experiences such as malaise (i.e., conditioned flavor avoidance), pleasure, or the receipt of energy (conditioned flavor preference). Hence, a negative experience with a sweet food, such as food poisoning, might lead to avoidance of that specific food rather than all sources of energy from foods with sugar [118]. By contrast, taste identifies the presence and quantity of potential nutrients or toxins without requiring learning. This is advantageous because it would be perilous to have to learn that bitter should be rejected and sweet accepted. Thus, taste and accompanying subjective experience (e.g., sweetness and liking) are critical for initial decisions about accepting or rejecting a food [119]. Both taste and flavor are able to evoke subjective affective experiences to enable fast, direct associations between actions (consumption) and outcomes prior to the generation of post-oral rewarding signals (shaping). Oral sensations and their accompanying subjective experiences thus occupy an intermediary position as proxy rewards between conditioned reinforcers and primary reward signals that are subliminal and delayed (see Figure 2B in the main text).

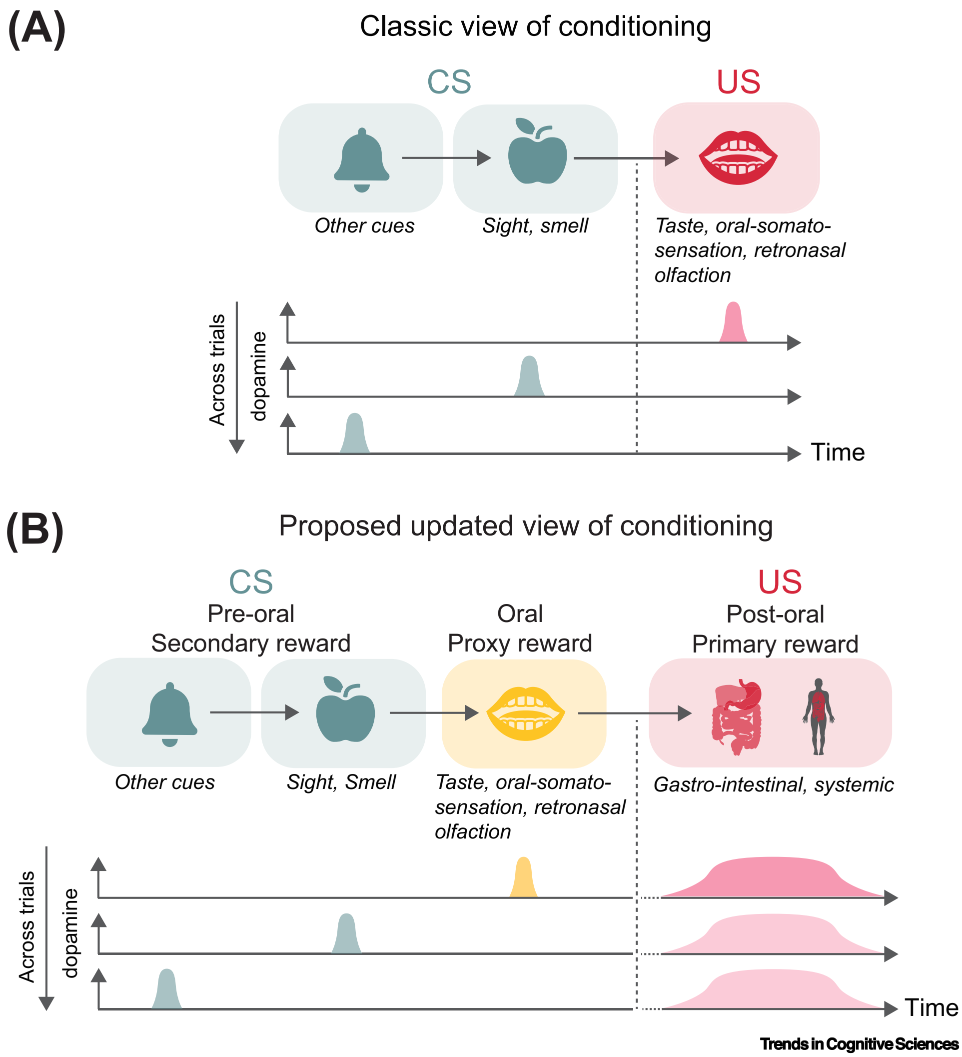

Figure 2. Expanded view of conditioning including post-oral primary reward signals. (A) In the classic view, the receipt of a primary reward (consumption of food or juice) triggers a phasic activation of midbrain dopamine neurons in response to the unconditioned stimulus (US) (red peak). When reward delivery is consistently preceded by a cue, such as a sound or visual signal [conditioned stimulus (CS)], the phasic dopamine response gradually shifts across trials from the moment of consumption to the earlier occurrence of the predictive cue, now considered a secondary reward (green peaks). (B) Proposed updated view of classical conditioning: Recent research suggests that the critical reinforcing signals from food or water are not primarily linked to oral sensory signals during consumption. Instead, primary reward signals are associated with post-oral processes that occur during digestion and absorption (red peak). These post-oral signals occur within a broad time window to accommodate variability in the timing and duration of those events. In this model, the immediate responses to oral signals act as proxy rewards (yellow), providing an early 'affective draft' of the value (delayed outcome) of the consumed food. These proxy rewards are distinct from pre-oral cues (secondary rewards), which, through conditioning, become associated with and predictive of the primary rewards. It remains an open question whether post-oral primary reward signals diminish over time as earlier signals increasingly predict their occurrence, as depicted by the lighter shading of the red peak.

actions and outcomes and solve the credit assignment problem [42]. A proxy reward signal linked to sweet taste initializes a food value, which accelerates learning the association of the action with the later energy when glucose is metabolized. This idea of using proxy reward signals to speed up learning is known as 'shaping' in both animal learning and models of RL [50,51].

Such shaping signals are useful to the degree that they reliably predict delayed consequences. Indeed, research in macaque monkeys demonstrated that trial-by-trial RL depends on immediate, oral-sensory food properties that indicate a food's nutrient composition [46,52]. This highlights a tight link between proxy reward signals and metabolic outcomes, although future research is needed to understand the interactions between early and late reward signals. The question whether agents can learn useful shaping signals from experience is an area of ongoing theoretical research [17], with implications for biological and artificial intelligence, particularly in designing artificial agents capable of navigating environments with sparse or delayed rewards [50,51,53].

Distinct striatal architectures for primary and proxy rewards

Emerging evidence suggests that oral proxy reward signals and post-oral primary reward signals engage distinct dopamine pathways. Early oral reward signals predominantly activate the ventral striatum (VS), particularly the nucleus accumbens, whereas post-oral signals primarily target the DS [41,54].

This anatomical distinction between early and late signals likely maps onto functional differences [55]: the VS, involved in reward anticipation and value learning, is associated with the hedonic aspects of food consumption [34,54], consistent with the proposed role of proxy reward signals. In contrast, the DS, linked to action learning and motivation, is consistent with the notion that post-oral primary reward signals serve as direct reinforcers of behavior [54]. Supporting this, early studies found that nigrostriatal dopamine depletion impaired feeding [56] and that restoring dopamine signaling in the DS, but not the VS, was sufficient to rescue feeding behavior and prevent starvation in dopamine-deficient mice [57]. More recent work shows that post-absorptive dopamine signaling in the DS is necessary for RL beyond mere motor function restoration [41]. Human positron emission tomography studies corroborate this distinction, showing that dopamine release during oral stimulation occurs in the VS (alongside the insula and hypothalamus), whereas post-meal dopamine responses are observed in the DS and basolateral amygdala [47,58]. This extensive network of responses highlights the importance of a broader network in RL that further shapes reward valuation, decision-making, and adaptive behavior [55].

Beyond credit assignment: early signals drive prospective control

Predictive pre- and intraoral signals do more than aid credit assignment; they enable organisms to anticipate future bodily states and act prospectively [59]. For instance, these cues help regulate food and water intake by forecasting expected hydration and energy levels without having to wait for delayed postingestive feedback.

Evidence for prospective control can be observed in increased insulin release and suppression of Agouti-related peptide expressing (AgRP) 'hunger' neurons in response to food-predicting cues, enhancing metabolism and directly modulating food consumption [60,61]. This mechanism can influence the final reward signal itself; insulin affects glucose conversion into energy, altering the post-oral reward signal from glucose oxidation. For example, it has been demonstrated that DIT following carbohydrate intake is modulated by the sweetness of the consumed beverage [35].

Predictive cues also enhance motivation and perception. This is evident in **cue-potentiated feeding**, which can occur even without hunger [62], and in the stimulation of cravings through mental simulation, such as imagining the taste of food [63]. In humans, responses to food cues predict cue-potentiated feeding and long-term weight gain [63]. Moreover, pre-consumption expectations about satiety significantly impact subjective fullness even hours later [64].

Thus, although primary reward signals reflect essential physiological variables (i.e., extractable energy from food), this energy availability is shaped by anticipatory and preparatory actions (i.e., metabolic adjustments). This highlights a closed loop where reward anticipation drives preparatory actions (allostasis), which influence post-oral feedback that refines future predictions.

Internal states, drives, and primary rewards

Primary reward signals for food and water are internally generated and driven by specific bodily events, such as nutrient detection (Figure 1). Yet, how valuable an outcome is depends not only on its nutritive content but also on individual needs, desires, goals, and context. For instance, a burger will be more valuable to a hungry person than a sated person (internal state/need), whereas the reward from consuming chocolate often seems independent of physiological need. What, then, is the relationship between primary reinforcing signals for sugar, fat, and water and internal states and drives?

Although the state dependency of early pre- and intra-oral signals is well-studied [25,55,65], precisely how internal states impact primary post-oral reinforcing signals is a field of active investigation. Available data suggest two types of interactions (Figure 3A): **state-driven primary reward signals**, whose occurrence is causally linked to a beneficial change in current internal state, and **event-driven primary reward signals**, elicited by appetition events. In both cases, reinforcement is amplified by internal **drive**, which reflects anticipated future imbalance (Figure 3B).

State-driven primary reward signals

Fluid consumption is a prime example where primary reward signals directly reflect the detection of an internal state change (Figure 3A). Here, a subset of VTA neurons has been identified that respond to changes in systemic osmolarity and are both necessary and sufficient for reinforcing fluid consumption [41]. Crucially, this dopamine response is driven by GABAergic neurons in the lateral hypothalamus (LH), which track fluid balance, aligning with prior research linking the hypothalamus to reinforcement and motivation [66–68]. Importantly, this indicates that the primary reward from water consumption is triggered by detecting an actual improvement in systemic hydration (fluid balance) – the redressing of an imbalance. This is different from a change in internal drive, which additionally depends on a desired and anticipated state and is represented by thirst-promoting neurons in the subfornical organ (SFO) (Box 3, Figure 3B). However, internal drive amplifies the primary reward signal during rehydration, consistent with the gating of reward signals by need state [22,25,65] and reminiscent of how physiological states gate sensory signals related to food [69].

Event-driven primary reward signals

Unlike water, primary reward signals for food seem to reflect 'absolute' direct signals caused by a concrete physiological event (i.e., glucose oxidation; see 'The case of sugar'; Figure 1) rather than a beneficial change in internal state [22,23] – appetition rather than satiation ([70], Figure 3A). This is supported by evidence that sugar is reinforcing beyond satiation [61] and that preferences can be formed for flavors even in sated animals [70].

Similar to thirst-promoting neurons in water reward, hunger-promoting AgRP neurons potentiate postoral VTA dopamine activity in response to food [41,67,71,72]. Generally, there is evidence for multiple different mechanisms by which internal states (i.e., hunger and satiety) modulate primary reward signals, including the sensing of gastric distension, hypothalamic nutrient sensing, and ghrelin and leptin-sensitive receptors on midbrain dopamine neurons [25,73,74]. In the case of overeating, negative feedback signals such as discomfort, pain, or nausea may compete with primary reward-based responses, without preventing primary rewards; however, more research

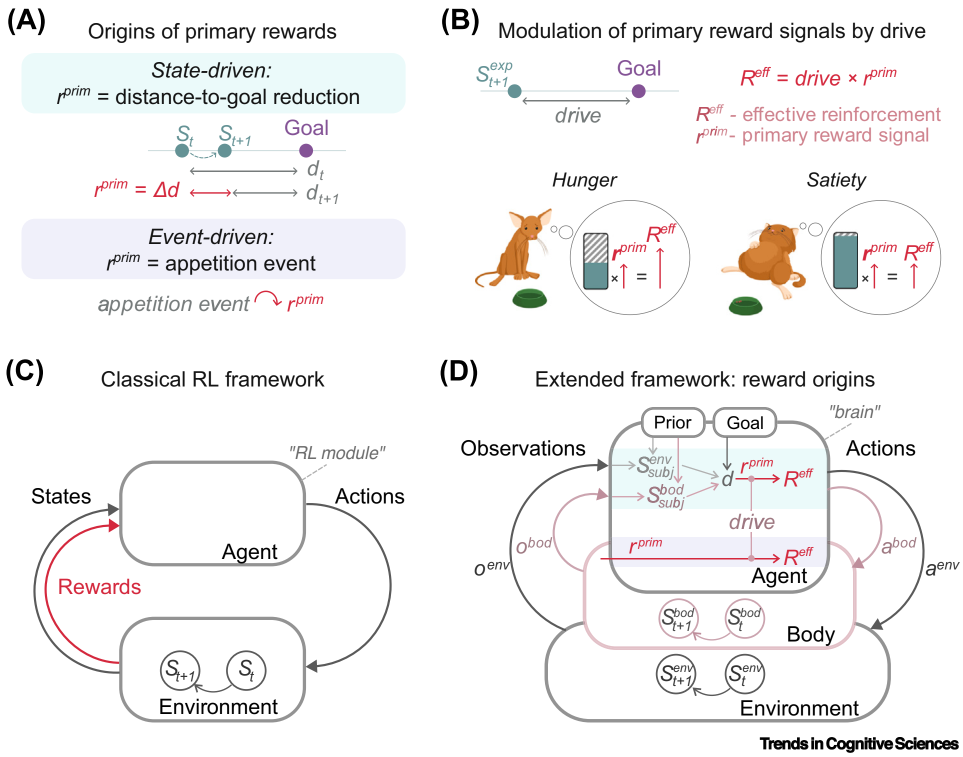

Figure 3. Extending the reinforcement learning framework to model internally generated primary rewards. (A) Reward generation mechanisms. Primary reward signals r^{prim} arise either from state deficits ('state-generated') or from appetition events ('event-generated'). In state-generated rewards, an agent compares its subjective state estimate to an internal goal; outcomes that reduce the distance d to that goal yield a reward. In event-generated rewards, specific physiological events (e.g., glucose → ATP conversion) directly elicit a reward. (B) Reward modulation by drive. Both reward types are boosted by anticipated deficits. Drive neurons (e.g.AgRP, SFO neurons) represent the expected future distance to a goal state, using predictive internal and external cues [102], so that reinforcement is more effective when deprivation is expected. (C) Traditional reinforcement learning (RL) framework. Agents select actions that alter the state of the environment (S_t) and receive scalar valenced reward signals, which update value estimates of expected cumulative future rewards (not shown; but see Box 1). Note that 'agent' denotes the RL module and 'environment' includes the external world and the rest of the organism, including the organisms' internal reward generation process. This important nuance is often overlooked when the framework is applied to biological agents and rewards are externally defined. (D) Extended RL framework. Accounting for primary reward signals: (1) Conceptualizing the body as a specific environment that provides noisy sensory observations (o^{bod}) about internal states (S^{bod}), which are affected by external (a^{env}) and internal actions (a^{bod}). From these observations and prior beliefs, agents form subjective state estimates ($S^{bod}{}_{subj}$) and compare them to internal goals (or set points) to represent the distance d. Additionally, they use cues and a predictive model to represent the expected future distance (*drive*). (2) Acknowledging that all reward signals are internally generated – there is no valenced signal from the external environment – instead, valence is generated within the agent in response to (neutral) incoming sensory observations. Rewards derive from subjective state evaluations or specific physiological events (A) and are amplified by expected deprivation (drive, B). Abbreviations: *a*, action; *bod*, body; *d*, distance; *env*, environment; *exp*, expected; *ext*, external; *o*, observation; R^{eff}, effective reward/reinforcement; r^{prim}, primary reward; *S*, state; *subj*, subjective.

is needed to clarify their interaction. In sum, internal need states (e.g., hunger) modulate but do not directly cause primary food reward signals (Figure 3B).

Does storage explain different architectures for different resources?

One potential explanation for these distinct architectures may be biological storage capacities: While the surplus energy from food can be stored longterm as fat for later use, water storage capacities are restricted, forcing the maintenance of osmolarity levels in a narrower range. This fundamental distinction likely influences neural circuits that regulate the reinforcement of behaviors

Box 3. The modulatory role of drives in reinforcement

The idea that behavior is reinforced to the extent that it brings an organism's internal states closer to their optimal or desired value goes back to drive reduction theory of motivation [122]. According to this theory, deviations of critical internal variables – such as energy levels, temperature, and fluid balance – from their desired values (referred to as 'set points' or 'set ranges,' which support the organism's functioning) create drives. These drives are need states that motivate the organism to seek out resources such as food or shelter. According to this theory, an outcome is rewarding and an action reinforced to the extent that it changes internal states of that organism to successfully reduce this homeostatic drive [87].

Thus, drive neurons must represent need states and be linked to the motivation to reduce drive [44,123–126]. For example, hunger-promoting AgRP neurons and thirst-promoting SFO in the hypothalamus fulfill these criteria [60,61,102,127]. Interestingly, stimulation of neither 'thirst' nor 'hunger' neurons drives midbrain dopamine activity or reinforcement directly [41], but instead they modulate the dopamine response to primary rewards [25,67,74,127].

This discrepancy may be explained by a recent shift in the understanding of what is represented by thirst and hunger neuron activity [44,60,61,102]. Instead of responding to actual caloric or fluid deficits, thirst and hunger neurons are increasingly believed to encode expected future states [102]. For example, their activity is rapidly inhibited upon the presentation of predictive food- or water-predicting cues, even prior to the actual detection of a state change [44,60,61,128]. These cue-elicited responses are shaped by variables such as food and water availability, palatability and caloric content, suggesting that these hunger and thirst neurons motivate behavior in an anticipatory fashion, consistent with an allostatic (rather than a homeostatic) model [103].

The anticipatory nature of their responses may explaining why a change in hunger or thirst neuron activity may not be a suitable source of post-oral reward signals: instead, a separate, 'ground-truth' account of actual change in energy or fluid balance is required – which serves both as a primary reward signal and as corrective feedback to update predictions about expected future states from hunger or thirst neurons [72,129].

acquiring these resources. Specifically, nonstorable resources (e.g., water) should be prioritized when needed and consumed in proportion to their restoring effect [75–77]. In contrast, storable resources (e.g., food) can be pursued beyond immediate need, remaining rewarding even after satiation – until excessive intake leads to discomfort or health risks due to overeating. Ideally, the neural architecture for such resources maintains a relatively stable reward signal that is heightened during deficits and moderated by long-term negative feedback (e.g., stomach fullness, body weight sensors that regulate body fat mass) to prevent overaccumulation [78,79]. Other factors, such as body composition and metabolic status associated with obesity, a chronic high-fat diet, or metabolic diseases, may additionally modulate primary reward processing by altering vagal sensitivity and dopamine signaling in response to nutrients [39,80].

Similar mechanisms may apply to more symbolic reinforcers used to obtain primary rewards that also have the capacity of storage. For instance, although there is no specific sensory channel or metabolic signal for processing money, it serves as a powerful reinforcer for humans – almost as potent as food or water rewards – and it can be stored [81–84]. Thus, if storage capacity shapes cognitive architectures for state-dependent reinforcement, monetary rewards (and other symbolic reinforcers) may engage neural circuits overlapping with food but not water rewards (see Outstanding questions).

Extending the reinforcement learning framework

Despite the success of RL for studying animal behavior, one of its core concepts, the very definition of reward, remains surprisingly paradoxical when applied to natural intelligence [12]: reward is treated as an external, objective, and stable quantity (Figure 3C, Box 1). This externally generated information serves as an ultimate 'ground-truth' correction signal upon which the agent can update their subjective beliefs about the value of states and actions. This makes sense in artificial agents, whose algorithms are restricted to the reward-based learning component and whose goals and reward functions are engineered. In contrast, for understanding natural intelligence, we need to study all components, including how reward signals are generated on the basis of internal psychological or physiological signals and states. Critically, RL and reward generation interact – meaning, in biological intelligence, that the origin of reward matters. So, what do the primary reinforcing signals for sugar, fat, and water tell us about reward generation mechanisms in the brain?

Multiple primary reward signals originate in the body

Primary reward signals in biological agents are not externally generated, and they do not track a single quantity of interest (e.g., calories). Instead, they originate in the body – the *milieu intérieur* [85] – which should therefore explicitly feature in any formal account of primary reward generation (Figure 3D). Because survival depends on acquiring energy, hydration, and avoiding harm, success is evaluated by monitoring their related bodily signals. Importantly, nutrient-specific rewards are transmitted through distinct but parallel gut–brain circuits (Figure 1) [38], and they interact supra-additively to shape food value [38,40,63]. Modular RL architectures that incorporate these types of multiple independent, resource-specific reward signals have been shown to outperform monolithic approaches that merge resources into a single scalar signal [86–88]. This paradigm shift aligns with research on heterogeneity of dopamine responses, which has led to theoretical updates in RL where scalar reward prediction errors (RPEs) are replaced by multidimensional RPEs [89–93].

Internal states are inferred and subjective

Incorporating the body as part of the agent's environment highlights that internal bodily states, like environmental states, are not directly observable but must be inferred from noisy visceral signals and prior beliefs – a process known as **interoception** [94–97] (Figure 3D). Interoception underpins effective regulation of the physiological variables critical for survival [97–100] and has been suggested to form the basis for emotional valence [101].

Crucially, information about current bodily states is combined with cues from the environment and learned associations to provide an estimate of future internal state in the service of prospective control [97,102–104]. As reviewed in 'Internal states, drives, and primary rewards,' the post-oral reinforcing signals for sugar, fat, and water rewards are indeed influenced by central representations and expectations of the state of the body. This is evident in GABAergic LH neurons tracking systemic hydration status and SFO neurons and AgRP neurons coding internal thirst and hunger drives (Box 3). Stimulating these regions was sufficient to modulate DA reward signals and behavior, mimicking natural dehydration or hunger [20,41].

Thus, interoceptive (subjective) estimates of internal state, rather than actual physiological changes, are responsible for the state dependency of reinforcement. Consequently, when formalizing reward generation mechanisms in an extended RL framework, internally generated reward signals should be tightly linked with subjective, internal states (Figure 3D).

Beyond homeostatic reinforcement learning

A major attempt at formalizing reward generation mechanisms in biological RL extends the concept of drive reduction to reward (Box 3) [87]. In this framework, utility arises from changes in internal state relative to a set point (desired state), where movement toward the set point is rewarding and deviation is punishing. Modern formulations of this idea incorporate prospective control (acting to avoid future drives) as well as subjective inference on the current state [87,97,98,104,105] or replace the 'landscape of (physiological) needs' with an emotional or affective landscape [106].

The recent literature reviewed here highlights two aspects of reward generation that have been overlooked by existing accounts. First, not all primary rewards stem from changes in internal state or drive. The event-driven reward signals for sugar and fat suggest that some resources (likely the ones that are storable) drive reinforcement even in the absence of current needs [107] (Figure 3A), although their effects are clearly amplified in states of deprivation (Figure 3B). Future work should determine whether such appetition signals are indeed tied to the capacity for storage, in which case, these seemingly 'hedonistic' drivers of reinforcement may actually reflect homeostatic regulation over longer timescales, likely with a predictive component (allostasis) [70].

Second, delayed and retroactive influences play a crucial role in reward computation. Post-digestive nutrient sensing can retroactively adjust the reward value assigned to previously consumed food, a process that standard RL models and even some homeostatic frameworks struggle to accommodate. This distinction points to a more flexible, temporally extended reward architecture for biological systems, where shaping signals or proxy rewards provide predictive estimates of utility, whereas interoceptive assessments of the impact of an outcome on immediate and future homeostasis serve as ground-truth feedback signals.

Goal-dependent rewards

Primary rewards are defined by the physiological benefits they offer to the organism. The fundamental goals implied by primary reward functions of biological agents are those of survival and reproduction. However, these broad goals translate into more specific subgoals, shaping behaviors around acquiring resources and avoiding threats on the basis of the agent's environment [108,109]. Understanding how agents update their reward functions in response to changing internal and external conditions is key to explaining complex, flexible behavior, particularly in intelligent agents, who set their own goals and experience reward upon achieving them [1,110].

But how are reward functions transformed to achieve cognitive goals that are more abstract or temporally distant? One possibility is that goal-directed behavior beyond primary rewards evolved by repurposing existing circuits for state-dependent primary reinforcement. In particular, the concept of drive reduction, originally applied to physiological needs, has since been translated to the cognitive domain through the idea of 'cognitive set points' [111]. In a remarkable parallel to physiological drive reduction, these models posit that value signals during the pursuit of more abstract cognitive goals may emerge from the comparison between a current state, the desired (goal) state, and the degree to which an action reduces the distance between them [12].

Others propose that internal goals actively shape an organism's motivation to pursue them [112–114], although how goals are selected and prioritized remains unclear. Despite different interpretations, these frameworks converge on a common theme: if agents generate reward signals on the basis of internal state changes, then they are also inherently capable of goal-dependent valuation.

Concluding remarks

This paper has reviewed research on ingestive behavior to uncover the nature and origins of biological reward functions. Traditionally, rewards have been linked to the immediate outcomes of actions – such as the sweet taste of biting into a juicy apple. However, it is now clear that the critical reinforcing signals from food and water do not stem directly from consumption. Instead, they arise from delayed post-oral primary reward signals during digestion and absorption, tracking vital resources for survival, such as nutrients, energy, and hydration.

These primary reward signals are connected to secondary and proxy rewards, such as cues and appetitive responses to taste, and they are modulated by an organism's internal states and goals.

Outstanding questions

Beyond the main macronutrients (fat, carbohydrate, protein), what additional micronutrients are involved in producing postoral signals?

What are the underlying primary reward signals associated with other physiologically relevant processes, such as sex, temperature regulation, and breathing?

How do these interoceptive reward signals become integrated with brain circuits supporting decision-making?

To what extent do interoceptive reward mechanisms generalize to intrinsic rewards related to curiosity, goal achievement, or novelty?

Does the ability to store energy from food provide organisms with the capacity to reinforce behavior beyond internal states? Could that provide additional insights into why overeating is common but overdrinking is rare, as well as mechanisms of addiction more generally?

Do storable secondary reinforcers (e.g., money) share neural architectures with energy-related rewards, but not with unstorable rewards, such as those linked to water intake?

How do other interoceptive signals, such as the processing of cardiac and breathing-related information, interact with reward systems?

How are the different dopamine signals for delayed internal and immediate external reinforcing signals integrated (e.g., within the dorsal and ventral striatum)?

How are multiple independently generated reward signals combined to produce a scalar feedback signal used for RL (or do they at all)?

What are the mechanisms by which delayed primary reward signal update early (predictive) reward responses upon consumption (e.g., flavor nutrient conditioning)?

On the other hand, what are the mechanisms by which prospective physiological changes at the time of consumption influence the strength of the delayed reinforcing signal?

These insights suggest an extended view of the role of reward in RL that shifts the focus from immediate sensory gratification to a more nuanced, internally generated, state-dependent reward system. Understanding these mechanisms not only clarifies how biological needs drive behavior but also opens new avenues for studying the interplay between primary, secondary, and proxy rewards (see Outstanding questions).

How does cognition shape the biological reward function?

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Declaration of interests

The authors have no interests to declare.

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